

A Model of Currency Linkages During Crises

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1 Abstract

This paper addresses two important questions in foreign exchange markets during crises: (1) whether cointegration relationships among major currencies persist across distinct crisis episodes, and (2) whether long-horizon currency returns show increased correlation as currency autonomy declines. We analyze daily spot rates for five major currency pairs (EUR/USD, USD/JPY, GBP/USD, USD/CHF, AUD/USD), focusing on the Global Financial Crisis (2007–2009) and the COVID-19 pandemic (2020–2022).

Our findings confirm existence of cointegration among major currencies during the Global Financial Crisis, consistent with reduced currency autonomy driven by common global risk factors. In contrast, the COVID-19 period exhibits a breakdown of long-run cointegration despite elevated volatility, driven by the emergence of a new, distinct global stochastic trend which increases the number of stochastic trends and erodes long-run linkages, reflecting greater currency-specific autonomy. These results challenge the standard crisis-contagion narrative by demonstrating that global crises do not uniformly increase FX integration. We highlight the importance of regime-dependent pass-through parameters in exchange rate models, discuss the relevant implications and explain the condition under which cointegration can be preserved across crisis periods.

2 Introduction

This paper investigates how global crises affect the long-run relationships among major currencies. Specifically, we address two questions: (1) Are cointegration relationships among currencies preserved during systemic crises? (2) How do long-horizon currency return correlations evolve as currency autonomy varies across market regimes?

Foreign exchange (FX) markets are the largest and most liquid over-the-counter financial markets, central to global trade, investment, and monetary transmission. Understanding the integration and comovement of currencies is vital for portfolio diversification, risk management, and tracing financial shock propagation. Episodes of systemic stress—most notably the Global Financial Crisis (GFC) of 2007–2009 and the COVID-19 pandemic—offer natural experiments to examine shifts in both short-run currency comovements and their long-run equilibrium relationships.

Under frictionless, integrated markets, asset pricing theory predicts that bilateral exchange rates share a common global stochastic discount factor (SDF), implying cointegration with rank $N - 1$ in an N -currency system (5; 9). However, structural frictions, heterogeneous monetary regimes, and crisis conditions can alter these long-run linkages. Empirically, the GFC—characterized by USD funding stress—strengthened common global factor influences and cointegration, consistent with theory. In contrast, the COVID-19 period reveals a surprising breakdown of cointegration, suggesting a rise in currency-specific stochastic trends and autonomy that challenges conventional crisis integration narratives.

Our contribution is to unify factor-driven exchange rate models, no-arbitrage asset pricing, and Johansen cointegration analysis into a regime-dependent framework that allows for time-varying currency autonomy. We introduce an autonomy parameter $\psi_{i,r}$ governing the exposure of currency i to global factors in regime r . Crises like the GFC compress $\psi_{i,r}$ toward zero, reflecting diminished autonomy and tighter long-run linkages; by contrast, the COVID-19 crisis corresponds to increased autonomy and the emergence of new, independent stochastic trends, which fragment the cointegrating structure. This framework offers a structural explanation for the observed shifts in cointegration rank and currency comovement patterns.

Our stylized model links $\psi_{i,r}$ directly to the cointegration rank, adjustment speeds, and long-run correlation structures estimated from vector error correction models (VECMs). Elevated global factor volatility and reduced autonomy yield higher long-run correlations and stronger network connectivity (??), whereas increased autonomy leads to weaker cointegration and greater heterogeneity across currencies—empirically confirmed during COVID-19.

Our work bridges structural asset-pricing theory with empirical cointegration analysis, applying this synthesis to both major and emerging FX markets across multiple crises. While

prior research emphasizes short-run volatility spillovers (20?) or regional currency blocs (17), we provide a comprehensive long-run equilibrium perspective with regime-dependent factor loadings and network diagnostics. Moreover, we document and structurally interpret the COVID-19-induced breakdown of long-run parity—a novel phenomenon absent in prior crises and largely unexamined in cointegration literature.

Our framework distinguishes between the existence of long-run equilibrium relationships (cointegration) and the degree of long-horizon return comovement. While cointegration imposes a structural condition on the number of independent stochastic trends, long-horizon return correlation reflects how tightly currencies co-move over extended horizons—even in the absence of cointegration. This distinction allows us to show that COVID-19 weakened structural linkages (cointegration), yet certain currencies continued to exhibit substantial long-horizon comovement, pointing to a regime where currencies load differently—though not independently—on evolving global risk factors.

Our analysis uncovers a novel structural phenomenon in the behavior of major foreign exchange markets during global crises. While prior research largely documents increased integration or temporary disruptions in cointegration during episodes such as the Global Financial Crisis, we find that the COVID-19 pandemic introduced a new global stochastic trend factor—distinct and independent from pre-existing common risk factors. This emergence increased the number of independent stochastic trends in the system, thereby eroding the long-run cointegration relationships among major currencies.

Crucially, we show that the presence of an additional stochastic trend does not necessarily weaken or destroy cointegration if the new trend is spanned by existing global factors. However, the COVID-19 crisis was characterized by a truly novel factor that could not be represented as a linear combination of prior global factors. This structural break led to the observed breakdown of long-run linkages, signaling a regime shift in the underlying global risk environment.

This finding challenges the prevailing crisis-contagion narrative, which assumes crises uniformly compress currency autonomy and strengthen long-run ties. Instead, our results highlight that crises can reshape the fundamental factor structure of currency markets in qualitatively different ways, with significant implications for asset pricing, risk management, and international portfolio diversification.

Previous studies on foreign exchange market integration during crises have largely focused

on shifts in short-run volatility spillovers (Melvin and Taylor, 2009; Aloui et al., 2014) or the temporary breakdown and subsequent restoration of long-run parity (Flood and Rose, 1995; Dungey et al., 2005). While these works document changes in cointegration rank or correlation patterns, they do not explicitly characterize the factor structure changes underpinning these shifts.

Our contribution lies in combining a structural asset pricing framework with cointegration analysis to demonstrate that a crisis can introduce genuinely new global risk factors, thereby increasing the number of independent stochastic trends. This insight deepens the understanding of how crisis regimes can lead to permanent structural breaks rather than transient deviations in exchange rate dynamics.

To our knowledge, this is the first paper to document and interpret the breakdown of long-run parity among major currencies during COVID-19 as the result of an emergent, distinct global stochastic trend. This enriches the literature by providing a mechanistic explanation for the observed heterogeneity in FX market responses to different global crises.

Why Focus on Foreign Exchange Markets? The foreign exchange (FX) market is uniquely suited to study long-run cointegration and crisis dynamics for several reasons. First, FX markets are the largest and most liquid financial markets globally, with daily turnover exceeding that of equity and bond markets combined, ensuring that prices reflect real-time global information with minimal frictions. Second, unlike equities and bonds, FX rates are inherently cross-border relative prices directly tied to macroeconomic fundamentals, monetary policy, and risk premia across countries. This relative pricing structure creates a natural environment to test long-run equilibrium relationships and cointegration, grounded in no-arbitrage and uncovered interest parity conditions, which are less cleanly applicable to segmented or jurisdiction-bound markets such as equities or sovereign bonds. Third, FX markets operate continuously across global time zones, enabling comprehensive high-frequency data coverage necessary to detect subtle regime shifts and structural changes, especially during crises. Finally, the dominant role of the U.S. dollar as a global funding currency amplifies the transmission of global risk shocks, making FX markets a natural laboratory to investigate the interplay between global systemic factors and currency-specific autonomy during extreme events. Together, these features establish FX markets as an ideal setting to study regime-dependent cointegration and crisis-induced market dynamics, offering insights that are less readily obtained from equity or bond markets. Let me know if you want me to adjust it to fit a particular section or style!

3 Literature Review

Long-run comovement and cointegration in FX

A foundational strand of the foreign exchange (FX) literature documents common stochastic trends among major currency pairs, employing cointegration methods to study long-run parity relations. Early work establishes that several dollar exchange rates share persistent common drivers, motivating vector error correction models (VECMs) and rank tests to identify cointegration relationships (Baillie and Bollerslev, 1989; Johansen, 1988, 1991). Complementing this, asset-pricing perspectives reconcile the apparent near-random-walk behavior of exchange rates at short horizons with the relevance of fundamentals in present-value terms (Engel and West, 2005). These insights underpin the widespread use of Johansen’s framework to test whether N dollar exchange rates admit $N - 1$ cointegrating relations in integrated FX markets.

Risk-based and intermediary-based views of FX

Modern asset pricing models link exchange rates to priced risk factors and intermediary constraints. Currency returns load on common factors such as the “dollar” and “carry,” which help explain the cross-section of currency risk premia (Lustig, Roussanov, and Verdelhan, 2011). Meanwhile, intermediary asset pricing models emphasize that exchange rates also respond to dealers’ balance-sheet constraints and shifts in international liquidity; these frictions can alter currency comovement by repricing currencies and affecting funding conditions (Gabaix and Maggiori, 2015). These approaches provide structural channels—risk prices and funding constraints—through which global shocks synchronize currencies while allowing heterogeneity across countries and over time.

Crises, funding stress, and the global dollar system

Crisis episodes reveal how dollar funding conditions transmit globally through FX markets. The Global Financial Crisis (GFC) impaired USD funding markets, causing sharp deviations in FX swap pricing, heightened margins, and a shortage of dollars outside U.S. banks (Baba and Packer, 2009; McGuire and von Peter, 2009). These dislocations manifested as violations of covered interest parity (CIP), which subsequent research modeled as equilibrium phenomena linked to dealer leverage and regulatory constraints (Du, Tepper, and Verdelhan, 2018). These mechanisms imply that under stress, currencies may load more strongly and more similarly on the global dollar factor, potentially compressing long-run autonomy even

amid turbulent short-run dynamics.

COVID-19 as a distinct stress event

The COVID-19 pandemic triggered an extraordinary “dash for dollars,” with severe stress in FX swap markets and unprecedented policy interventions including central bank swap lines and liquidity facilities. BIS analyses document acute dislocations in March 2020 followed by normalization as swap lines expanded (Schrimpf, Shin, and Sushko, 2020). Related work shows that swap lines mitigated dollar funding shortages and narrowed CIP deviations across jurisdictions (Bahaj and Reis, 2020). While these studies emphasize short-run funding stresses, they raise important questions about whether COVID also altered *long-run* currency linkages in ways distinct from the GFC.

Structural breaks, regime dependence, and inference

From a time-series perspective, identifying regime shifts in cointegration relationships is critical. Residual-based tests permit endogenous single breaks in cointegrating vectors (Gregory and Hansen, 1996), while multiple-break frameworks support inference in high-dimensional systems (Qu and Perron, 2007). These tools complement Johansen’s cointegration tests to assess whether cointegration weakens, vanishes, or re-emerges across crisis regimes such as the GFC and COVID.

Connectedness and network approaches

A complementary literature employs variance-decomposition-based connectedness metrics (Diebold and Yilmaz, 2012) and network-theoretic methods (e.g., minimum spanning trees, PMFGs) to measure the strength and direction of cross-market spillovers. These approaches distinguish the *number* of common stochastic trends from the *tightness* of co-movement (covariance or connectedness), offering nuanced insights into crisis dynamics.

How this paper advances the literature

This paper contributes along four key dimensions:

(1) Unified Theory and Empirics Across Regimes

We embed a no-arbitrage, log-normal stochastic discount factor (SDF) with nonstationary global factors into a pass-through framework that directly maps to Johansen cointegration

and VECM objects. This framework nests the classic $N - 1$ cointegration benchmark and also accommodates multi-trend regimes where rank may decline or increase, capturing scenarios often missed by correlation- or short-run spillover-based analyses (cf. Johansen, 1991; Du, Tepper, and Verdelhan, 2018).

(2) Structural Interpretation of Autonomy

We introduce currency- and regime-specific autonomy parameters governing factor pass-through, providing a structural interpretation of cross-country heterogeneity highlighted in EM resilience studies (Aizenman and Jinjara, 2014) and observed comovement dynamics during both the GFC and COVID (Baba and Packer, 2009; Schrimpf, Shin, and Sushko, 2020).

(3) Distinguishing Rank from Tightness

We separate the *number* of common stochastic trends (cointegration rank) from the *tightness* of long-run co-movement (long-run covariance and connectedness). This distinction clarifies how a crisis can simultaneously exhibit high correlation yet *lower* cointegration rank if additional nonstationary global factors emerge—an insight supported by intermediary-based models and funding market evidence.

(4) COVID as a Structural Outlier

While the GFC reflects compressed autonomy and strengthened long-run cointegration, we document that the COVID-19 crisis coincided with weaker cointegration among major currencies. This is consistent with the emergence of new global stochastic trends, distinct funding mechanisms, and novel policy backstops (e.g., swap lines), thereby updating the crisis-integration paradigm in light of modern dollar funding architecture (Schrimpf, Shin, and Sushko, 2020; Bahaj and Reis, 2020).

Together, these contributions bridge classic cointegration literature (Baillie and Bollerslev, 1989; Johansen, 1988, 1991) with contemporary finance theory—incorporating risk factors, intermediary frictions, and policy interventions—to provide a structural account of why long-run FX linkages can strengthen during one crisis yet weaken in another.

4 Model Setup

We now present a regime-dependent factor model linking the long-run dynamics of N non-USD exchange rates to a set of nonstationary global factors. The U.S. is the numeraire. Let the spot exchange rate be

$$S_{i,t} = \text{units of currency } i \text{ per USD} \quad (i = 1, \dots, N),$$

so an increase in $S_{i,t}$ indicates USD appreciation against currency i . Define

$$X_{i,t} = \log S_{i,t}, \quad X_t = (X_{1,t}, \dots, X_{N,t})'.$$

Pricing Kernel (SDF)

We price assets in USD using a log-normal stochastic discount factor (SDF):

$$\log M_{t+1} = -r_t - \frac{1}{2} \varsigma_r^2 - \xi_{t+1}, \quad (1)$$

where r_t is the USD short rate, ξ_{t+1} is the SDF innovation, and $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$ by construction.

Multi-factor global structure

In regime r , the innovation admits q_r nonstationary global factors:

$$\xi_{t+1} = \lambda_r' g_{t+1} + v_{t+1}, \quad \mathbb{E}_t[g_{t+1} v_{t+1}] = 0, \quad (2)$$

where:

- $g_t = (g_{1,t}, \dots, g_{q_r,t})'$ are q_r independent $I(1)$ global factors, e.g., USD funding stress, global risk appetite, commodity-price shocks.
- $\lambda_r \in \mathbb{R}^{q_r}$ are SDF-level factor loadings in regime r .
- $v_{t+1} \sim \mathcal{N}(0, \sigma_v^2)$ is an idiosyncratic USD pricing-kernel shock orthogonal to g_{t+1} .

Each factor follows a random walk with regime-dependent drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_k,r} \varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1), \quad k = 1, \dots, q_r. \quad (3)$$

The total SDF innovation variance is:

$$\zeta_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_v^2, \quad \Sigma_{g,r} = \text{diag}(\sigma_{g_1,r}^2, \dots, \sigma_{g_{q_r},r}^2).$$

The currencies each have varying exposures to the vector of global factors (e.g., global risk, liquidity conditions, macroeconomic policy shifts), which means that the common-factor block loads differently on each currency: for some currencies, one or more factor loadings are large (high coefficients along specific factor directions), while for others they are small or negligible. In (??), the innovation to the stochastic discount factor is decomposed into two conceptually distinct components: a *systematic* term, $\lambda_r^\top \mathbf{g}_{t+1}$, and an *idiosyncratic* term, v_{t+1} . The q_r -dimensional vector $\mathbf{g}_{t+1}$ captures shocks that are common—though not necessarily equally important—to all currencies; examples include shifts in USD funding conditions, coordinated changes in major central bank policy stances, or global risk-on/risk-off episodes. The vector λ_r measures the sensitivity of the *pricing kernel* to each of these common shocks in regime r , and in the affine solution each currency i inherits its own q_r -vector of pass-through coefficients $\lambda_{i,r}$, reflecting its specific exposure profile to the different global factors. By contrast, v_{t+1} represents purely currency-specific noise in the SDF, orthogonal to $\mathbf{g}_{t+1}$ by construction. Its variance σ_v^2 fully determines its scale, so no additional factor loadings are needed at the SDF level. Economically, the vector λ_r captures how each orthogonal component of the common-factor block propagates through the pricing kernel, while the heterogeneous $\lambda_{i,r}$ map those common shocks into individual currency returns; the idiosyncratic term v_{t+1} is already asset-specific and its effects are entirely absorbed by the currency it directly influences.

Payoffs, No-Arbitrage and Affine Solution

A one-period claim pays $D_{i,t+1}$ units of currency i ; its USD price is

$$P_{i,t} = \mathbb{E}_t \left[M_{t+1} \frac{D_{i,t+1}}{S_{i,t+1}} \right]. \quad (4)$$

For a pure FX claim delivering one unit of currency i (i.e. $D_{i,t+1} = 1$), replication implies $P_{i,t} = 1/S_{i,t}$. No-arbitrage gives:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

or equivalently,

$$1 = \mathbb{E}_t \left[M_{t+1} \frac{S_{i,t}}{S_{i,t+1}} \right] = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right], \quad (5)$$

our key FX pricing restriction. The intuition behind the no-(riskless)-arbitrage condition is that the dollar price paid today for a future claim of one unit of foreign currency

$$\mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

must equal the current dollar cost of acquiring that same unit of foreign currency,

$$\frac{1}{S_{i,t}},$$

If this equality does not hold, an investor could exploit the difference to earn a profit without risk. Any profit that involves bearing risk is not considered a riskless arbitrage profit and is therefore outside the scope of our model.

Solving for Exchange-Rate Levels

Assuming (i) the SDF is log-normal as in (??)–(??), and (ii) fundamentals z_t ($K \times 1$) are stationary and independent of the global-factor innovations, the log-linear solution takes the affine form

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \quad (6)$$

with $u_{i,t}$ stationary, and

$$\mu_{i,r} := [C_t \mid r] = -\bar{r}_r - \frac{1}{2} \varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \kappa_{i,r}, \quad (7)$$

$$\kappa_{i,r} \equiv \frac{1}{2} [(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \Delta(\Gamma_i^\top z_t + u_{i,t})], \quad (8)$$

$$\Gamma'_i z_t = \sum_{k=1}^K \gamma_{i,k} z_{k,t}, \quad z_t = (z_{1,t}, \dots, z_{K,t})' \text{ is stationary}, \quad (9)$$

$$u_{i,t} = X_{i,t} - \mu_{i,r} - \lambda'_{i,r} g_t - \Gamma'_i z_t, \quad u_{i,t} \text{ stationary}. \quad (10)$$

Here $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$ is the regime- r mean short rate; $(\mu_r, \Sigma_{g,r}, \lambda_r)$ are regime parameters of the q_r -dimensional global-factor process and SDF; and σ_v^2 is the variance of the SDF idiosyncratic residual.

The global-factor vector g_t enters linearly in the pricing kernel, so $X_{i,t}$ must load linearly on g_t . Currencies differ in their exposure profiles to the q_r orthogonal global factors, with some exhibiting large loadings in certain factor directions and others having smaller or near-zero loadings. This heterogeneity is captured by the currency-specific vector $\lambda_{i,r}$, while λ_r measures the sensitivity of the pricing kernel to each global factor in regime r . In the SDF, the innovation decomposition

$$\xi_{t+1} = \lambda_r^\top g_{t+1} + v_{t+1}$$

separates the *systematic* component, $\lambda_r^\top g_{t+1}$, from the *idiosyncratic* component, v_{t+1} . The vector g_{t+1} represents shocks common to all currencies—such as shifts in global risk appetite, changes in major reserve currency funding conditions, or synchronized monetary policy moves—but whose economic impact varies across currencies according to their $\lambda_{i,r}$ profiles. By contrast, v_{t+1} is orthogonal to g_{t+1} by construction, representing purely currency-specific noise in the SDF; its variance σ_v^2 fully determines its scale, so no additional factor loadings are needed for this term at the SDF level.

The vector z_t is taken to be stationary so that exchange rates can load on *mean-reverting* conditions without introducing additional stochastic trends. While short-horizon exchange rates often approximate random walks—weakening their level relations with trending fundamentals¹—stationary variables capture time-varying but transitory forces that help explain cross-sectional heterogeneity and conditional dynamics.² Treating z_t as stationary preserves the common $I(1)$ trends driven solely by g_t and leaves the cointegration rank determined by the rank of Λ_r ; standard VECM/CVAR frameworks allow such stationary exogenous controls without affecting rank tests.³

Finally, the residual $u_{i,t}$ measures the deviation of $X_{i,t}$ from the level implied by its long-run relation with g_t and z_t . If $u_{i,t}$ were nonstationary, it would imply an additional persistent factor orthogonal to g_t , contradicting the assumption that g_t spans all nonstationary variation. Stationarity of $u_{i,t}$ ensures that any cointegration between $X_{i,t}$ and g_t is not impaired, and confines $u_{i,t}$ to mean-reverting, thereby preserving the internal consistency of the model’s long-run factor structure.

¹See Engel and West (2005) for the near-random-walk implication; classic out-of-sample evidence includes Meese and Rogoff (1983).

²Taylor-rule fundamentals and related stationary deviations have documented predictive content at certain horizons; see, e.g., Molodtsova and Papell (2009).

³See Johansen’s VECM framework as commonly implemented.

The USD price of one unit of currency i delivered at $t + 1$ is $1/S_{i,t}$. No-arbitrage implies:

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}].$$

Solving under log-normality and stationary fundamentals z_t yields:

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \quad (11)$$

where $\lambda_{i,r} \in \mathbb{R}^{q_r}$ reflects currency-specific exposure to the q_r global factors, $g_t \in \mathbb{R}^{q_r}$.

Regime-Dependent Pass-Through (“Autonomy”)

In order to model how the sensitivity of exchange rates to the global factor vector changes across regimes, we parameterize the factor loadings for currency i in regime r as

$$\lambda_{i,r} = (1 - \psi_{i,r}) \odot \bar{\lambda}_i, \quad \bar{\lambda}_i \neq 0 \text{ fixed}, \quad (12)$$

where $\psi_{i,r} = (\psi_{i1,r}, \dots, \psi_{iq_r,r})' \in [0, 1]^{q_r}$ is the vector of “autonomy” parameters of currency i from each global factor in regime r , $\bar{\lambda}_i \in \mathbb{R}^{q_r}$ is the baseline (regime-invariant) vector of factor loadings, and $\odot$ denotes the Hadamard (elementwise) product. A lower $\psi_{ik,r}$ implies stronger pass-through from factor k (larger $|\lambda_{ik,r}|$), meaning that the currency’s value is more sensitive to movements in that global factor. A higher $\psi_{ik,r}$ implies greater autonomy from factor k in that regime.

The following assumptions describe how volatility and autonomy evolve across regimes, and ensure the statistical tractability of the model:

- A1** For at least one factor k , $\sigma_{g_k, \text{crisis}}^2 > \sigma_{g_k, \text{pre}}^2$ and $\sigma_{g_k, \text{crisis}}^2 \geq \sigma_{g_k, \text{post}}^2$. Crisis periods are characterized by heightened volatility in one or more global factors, reflecting increased uncertainty, tighter global financial conditions, or both.
- A2** For all i, k , $\psi_{ik, \text{crisis}} \leq \psi_{ik, \text{pre}}$. Autonomy compresses during crises as market stress and funding pressures force more synchronized exchange-rate movements with the global factors.
- A3** For a non-empty set $\mathcal{E}$ (emerging markets) and for at least one factor k , $\psi_{ik, \text{post}} > \psi_{ik, \text{crisis}}$. Certain countries regain autonomy from one or more global factors after crises due to

policy adjustments, reserve accumulation, or reduced reliance on specific channels of external funding.

A4 z_t and $u_{i,t}$ are stationary and independent of the global-factor innovations ε_{t+1} . Stationarity of z_t ensures they capture mean-reverting fundamentals without introducing new stochastic trends; stationarity of $u_{i,t}$ preserves cointegration between $X_{i,t}$ and its drivers, while independence from ε_{t+1} maintains orthogonality between predictable components and shocks.

Factor-loading structure

For each regime r , define the $N \times q_r$ pass-through matrix

$$\Lambda_r = \begin{bmatrix} \lambda'_{1,r} \\ \vdots \\ \lambda'_{N,r} \end{bmatrix} = \begin{bmatrix} (1 - \psi_{11,r}) \bar{\lambda}_{11} & \dots & (1 - \psi_{1q_r,r}) \bar{\lambda}_{1q_r} \\ \vdots & \ddots & \vdots \\ (1 - \psi_{N1,r}) \bar{\lambda}_{N1} & \dots & (1 - \psi_{Nq_r,r}) \bar{\lambda}_{Nq_r} \end{bmatrix},$$

which stacks the currency-specific loading vectors $\lambda'_{i,r}$ across i .

Here:

- $\lambda_{i,r} \in \mathbb{R}^{q_r}$ are the currency-specific loadings on the q_r global factors in regime r .
- $\psi_{ik,r}$ is the “autonomy” of currency i from factor k in regime r ; lower values imply higher pass-through from that factor.
- Γ_i loads on stationary fundamentals z_t , which do not alter the cointegration rank.
- $u_{i,t}$ is stationary and mean-reverting.

Exchange-rate system

With these definitions, the cross-section of exchange rates in regime r is

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t, \tag{13}$$

where $\mu_r \in \mathbb{R}^N$ collects the regime-specific intercepts, $g_t \in \mathbb{R}^{q_r}$ is the vector of nonstationary global factors, Γ is the $N \times K$ matrix of stationary-factor loadings, and u_t is the vector of stationary residuals.

Cointegration Rank and Regime Dependence

Given the affine representation

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

the long-run dimension of comovement among the N exchange rates in regime r is determined by the rank of the pass-through matrix Λ_r .

The vector $g_t \in \mathbb{R}^{q_r}$ collects the q_r nonstationary global factors relevant in regime r , while $\Lambda_r \in \mathbb{R}^{N \times q_r}$ stacks the currency-specific loading vectors $\lambda'_{i,r}$. Stationary fundamentals z_t and residuals u_t do not affect the cointegration rank because they are $I(0)$.

The number of distinct common stochastic trends transmitted to X_t in regime r is

$$\text{rank}(\Lambda_r) \leq \min(N, q_r).$$

This yields the standard cointegration rank result:

- If $\text{rank}(\Lambda_r) = 1$, there is a *single* nonstationary common component, and the system has $N - 1$ cointegrating relations (the single-factor baseline).
- If $\text{rank}(\Lambda_r) = k > 1$, there are $N - k$ cointegrating relations (multiple independent global trends drive the system).
- If $\text{rank}(\Lambda_r) = N$, there is no cointegration (each currency contains its own stochastic trend).

Regime Variation

Both the factor volatilities $\sigma_{g_k,r}^2$ and the autonomy parameters $\psi_{ik,r}$ can vary across regimes, altering the effective rank of Λ_r . For example:

- A crisis may *increase* factor volatilities (making existing common trends more dominant) and/or *compress* autonomy parameters (aligning currencies more closely with the factors).
- If a new global factor emerges—e.g., a pandemic-specific risk driver during COVID—it can raise $\text{rank}(\Lambda_r)$ and reduce the number of cointegrating relations. In the extreme, if enough independent trends are introduced, cointegration may disappear even if it was present pre-crisis.

Key Regime Assumptions

We impose the following regime-dependent structure on volatilities and autonomy:

- A1 Crisis volatilities:** $\sigma_{g_k, \text{crisis}}^2 \geq \sigma_{g_k, \text{pre}}^2$ for all k , with strict inequality for at least one k . *Crisis periods exhibit heightened volatility in at least one global factor, reflecting increased uncertainty and tighter global financial conditions.*
- A2 Autonomy compression:** $\psi_{ik, \text{crisis}} \leq \psi_{ik, \text{pre}}$ for all i, k . *Market stress and funding pressures force more synchronized exchange-rate responses to the global factors.*
- A3 Post-crisis partial recovery:** For some emerging markets $i \in \mathcal{E}$ and some factors k , $\psi_{ik, \text{post}} > \psi_{ik, \text{crisis}}$. *These currencies regain some autonomy after crises through policy adjustments, reserve accumulation, or reduced reliance on specific funding channels.*
- A4 Stationarity and orthogonality:** z_t and u_t are stationary and orthogonal to factor innovations ε_{t+1} . *This preserves the cointegration rank implied by Λ_r and maintains orthogonality between predictable components and shocks.*

Main Propositions

The following propositions summarize the key theoretical implications of the multi-factor, regime-dependent pass-through model developed above. They connect the statistical properties of the pass-through matrix Λ_r to observable patterns of cointegration, long-run co-movement, and regime shifts in exchange-rate dynamics. Formal proofs are provided in the Appendix.

Proposition 1 (Rank–Cointegration Link)

Given regime r and assumptions A1–A4, the cointegration rank of X_t is

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

Economic interpretation: This result formalizes the direct link between the number of independent nonstationary global trends and the existence of long-run exchange-rate linkages. If $\text{rank}(\Lambda_r) = 1$, then all persistent variation in each exchange rate is driven by a single common global factor (e.g., the USD funding/liquidity factor), yielding $N - 1$ cointegrating relationships, meaning the exchange rates have established long run relationships and are related in the long run in $N - 1$ different ways, where in each case the residual is stationary, with the residual being the spreads between the exchange rate normalized on and a linear

combination of the other exchange rates. As $\text{rank}(\Lambda_r)$ increases, additional nonstationary drivers (for example such as stochastic factors like commodity-price trends, regional funding shocks, or segmented risk premia) introduce new sources of long-run divergence, reducing the number of cointegrating relations. In the extreme case $\text{rank}(\Lambda_r) = N$, each currency has its own independent stochastic trend, they do not share any common stochastic trend, and hence no long-run relationship survives as no combination of something that do not share anything in common would yield any established stable long run relationship.

Proposition 2 (Crisis Co-Movement Amplification of Long Horizon Currency Returns)

If $\text{rank}(\Lambda_r)$ remains unchanged in crisis, but (i) $\sigma_{g_k,r}^2$ rises for one or more factors, and (ii) $\lambda_{ik,r}$ is similar across currency for each global factor, i.e. react similarly to each global factor, which means there is convergence in response to the global factors across currencies (autonomy compression, each currency has less autonomy in its reaction to global factors), then long-run correlations, that is the correlation of long run currency returns across currencies increases, and factor volatility explains a larger share of the total variance of the long horizon returns for each currency.

Economic interpretation: Crises amplify co-movement through two reinforcing channels:

1. **Variance channel:** Higher volatility in the common factors increases their share of total long-horizon variance.
2. **Loading-alignment channel:** Lower dispersion in $\psi_{ik,r}$ makes currency responses more similar in magnitude and sign, reducing idiosyncratic offsets.

Even without introducing new stochastic trends, these channels produce tighter long-run alignment—a pattern consistent with the “risk-on/risk-off” behavior observed during global financial stress.

Proposition 3 (Rank Change in Crisis Lowers Cointegration Evidence)

If new persistent global factors emerge or autonomy shifts such that $\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}})$, the number of cointegrating relations falls. If $\text{rank}(\Lambda_{\text{crisis}}) = N$, cointegration disappears entirely.

Proposition 4 (Rank Change in Crisis)

If new **persistent global shocks** emerge, or if **autonomy shifts** such that

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

then the number of cointegrating relationships falls. In the extreme case where $\text{rank}(\Lambda_{\text{crisis}}) = N$, cointegration disappears entirely.

Economic interpretation: Certain crises—such as the COVID-19 shock—introduce structural changes that are *not spanned* by pre-crisis factors, such as pandemic-specific risk premia, supply-chain dislocations, or regime-specific commodity shocks.

If these shocks:

- are **persistent** (i.e., follow an $I(1)$ process),
- load **heterogeneously** across currencies, and
- are **linearly independent** of existing global factors,

then they increase the effective dimension of the global factor process g_t , raising the rank of Λ_r .

This leads to fewer cointegrating vectors, and in the limiting case of full rank ($\text{rank} = N$), a complete breakdown of long-run comovement across currencies.

Economic interpretation: Certain crises—such as the COVID-19 shock—introduce structural changes that are not spanned by pre-crisis factors: e.g., pandemic-specific risk premia, supply-chain dislocations, or regime-specific commodity shocks. If these shocks load persistently across currencies but in a pattern independent of pre-existing factors, they increase the effective dimension of g_t and the rank of Λ_r . The result is fewer cointegrating relations, with the extreme of rank N corresponding to a complete breakdown of long-run linkage.

Interpretation of Independent Stochastic Trends in Global Shocks

An **independent stochastic trend** is, by construction, a **new common stochastic trend** that *cannot be expressed as a linear combination of the existing trends*. This distinction is central to understanding structural changes in the cointegration properties of a multivariate system.

Specifically, an independent stochastic trend:

- **Increases the number of stochastic trends** driving the system, i.e., it raises the rank of the permanent (nonstationary) component;
- **Reduces the number of cointegrating relations**, since the cointegration dimension is $N - \text{rank}(\Lambda)$;
- And reflects the arrival of a **new persistent force** affecting multiple series in a way not previously spanned by existing global trends.

Hence, when we say that *COVID introduced an independent stochastic trend*, we mean:

- The shock was new and **common** across currencies (not idiosyncratic),
- It was **persistent** (not transitory or mean-reverting),
- And it was **non-redundant** — it could not be written as a combination of pre-existing trends, thereby altering the long-run structure of the system.

This leads to an *increase in the number of new common stochastic trends*, a reduction in cointegrating relations, and a potential *rank change* in the factor loading matrix Λ during episodes of global stress.

5 Data

Our dataset consists of daily spot exchange rates, covering recognized crisis periods in the global market. All exchange rates are quoted against the U.S. dollar (USD). Prices are transformed to natural logarithms for cointegrated analysis, and returns are calculated as first differences of log prices for the analysis of the correlation of long run currency returns. We focus on the most liquid “major” currency pairs, representing a blend of global reserve currencies, safe-haven assets, and commodity-linked units:

- **EUR/USD** — Euro, the second most important reserve currency and the deepest non-USD FX market.
- **GBP/USD** — British pound, historically significant but subject to geopolitical regime shifts (e.g., Brexit).
- **USD/JPY** — Japanese yen, a major safe-haven asset with persistent interest rate differentials.

- **USD/CHF** — Swiss franc, another canonical safe-haven currency.
- **AUD/USD** — Australian dollar, a commodity-linked currency influenced by terms-of-trade shocks.

Crisis Subperiods. To examine stability and regime dependence in long-run currency relationships, we split the sample into key subperiods:

1. **Global Financial Crisis (GFC):** August 2007 – June 2009.
2. **COVID-19 Crisis:** Core pandemic phase from February 2020 – December 2021, with extended windows (October 2019 – December 2022) to capture pre-crisis build-up and post-crisis adjustment.

These subperiods align with macro-financial chronology and allow direct comparison of cointegration stability across regimes.

TODO: PUT CHARTS OF THE DIFFERENT EXCHANGE RATES HERE IN THE TWO CRISIS PERIODS

6 Methodology

6.1 Unit Root Tests

We begin by verifying the order of integration of each exchange rate series using the **Augmented Dickey–Fuller (ADF)** and **Phillips–Perron (PP)** tests. Specifically:

- **Levels:** We include intercepts (and deterministic trends when appropriate) to test for unit roots and confirm nonstationarity $I(1)$.
- **First Differences:** We confirm stationarity $I(0)$ for returns.

This ensures that all series share the same integration order, a precondition for Johansen cointegration analysis. For both ADF and PP tests, the null hypothesis is the presence of a unit root (nonstationarity), while the alternative is stationarity. Tests are implemented in both levels and first differences, and the resulting statistics are reported below.

6.2 Johansen Cointegration

Having established the integration properties of the data, we apply the **Johansen (1988, 1991)** cointegration procedure to examine whether stable long-run equilibrium relationships exist among the $N = 5$ major currencies. The vector error correction representation is:

$$\Delta Y_t = \mu + \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t, \quad (14)$$

where Y_t is the $N \times 1$ vector of log spot rates, Γ_i capture short-run dynamics, and $\Pi = \alpha\beta'$ captures long-run equilibrium relations. Here, β contains the cointegrating vectors, while α are the speed-of-adjustment coefficients. If $r = \text{rank}(\Pi) = 0$, no long-run equilibrium exists; if $0 < r < N$, then r distinct cointegrating relationships characterize long-run dynamics.

Johansen proposes two likelihood-ratio statistics to determine r :

1. **The trace test**, which evaluates the null hypothesis of at most r cointegrating relations against the alternative of more than r :

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^N \ln(1 - \hat{\lambda}_i), \quad (15)$$

where T is the sample size and $\hat{\lambda}_i$ are the estimated eigenvalues (canonical correlations), ordered as $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_N$.

2. **The maximum eigenvalue test**, which evaluates the null of exactly r cointegrating relations against the alternative of $r + 1$:

$$\lambda_{\text{max}}(r, r + 1) = -T \ln(1 - \hat{\lambda}_{r+1}). \quad (16)$$

Both tests are widely used in empirical finance applications. However, Lütkepohl, Saikkonen, and Trenkler (2001) show that the trace statistic generally exhibits superior power in small and moderate samples. Since our segmented crisis-period datasets are not especially large, we therefore base most of our cointegration conclusions on the trace statistic.

Normalization in Johansen Tests. Johansen's procedure is invariant to the choice of normalization: selecting a variable to normalize on only affects presentation of the cointegrating vector, not the rank. In our analysis, we normalize on the euro (EUR/USD). The euro is the most liquid and least policy-managed non-USD currency, making it a natural benchmark for global FX linkages. Alternative numéraires (GBP, AUD, CHF, JPY) are more prone to idiosyncratic or regime-specific shocks, which could bias interpretation of the vectors.

6.3 Vector Error Correction Model (VECM)

When cointegration is established, we estimate a VECM to quantify the strength of the equilibrium relationships. The speed of adjustment coefficients in α measure how quickly deviations from long-run equilibrium are corrected. A larger (in absolute value) coefficient signals a stronger restoring force toward equilibrium, which we interpret as evidence of robust cointegration dynamics.

6.4 Rolling Return Correlations

To capture broader patterns of dependence that may arise even in the absence of cointegration (i.e., without common stochastic trends binding currencies together), we compute rolling returns for each currency over horizons of one day, one week, one month, and one quarter. We then examine correlations across the five currency pairs to assess the strength of their linkages at different horizons.

6.5 Average Pairwise Correlations

As a parsimonious measure of the overall degree of dependence among the five currencies, we compute the simple average of all pairwise correlations of their returns. Let $R_{i,t}$ denote the return of currency i at horizon h (where $i = 1, \dots, 5$), and let ρ_{ij} denote the sample correlation coefficient between $R_{i,t}$ and $R_{j,t}$ for $i \neq j$. The total number of distinct pairwise correlations is

$$\binom{5}{2} = 10.$$

The average correlation is then defined as

$$\bar{\rho} = \frac{1}{\binom{5}{2}} \sum_{i < j} \rho_{ij},$$

which corresponds to the mean off-diagonal element of the 5×5 correlation matrix of returns. A higher value of $\bar{\rho}$ indicates stronger average linkages among the currencies, consistent with greater market integration or contagion, while a lower value implies weaker connections and greater potential for diversification.

Although this measure averages across heterogeneous bilateral relationships, it provides a tractable single-index summary of systemic currency return dependence. We track $\bar{\rho}$ across horizons and crisis periods to assess how exchange rate co-movement evolves over time.

Complementarity of Approaches. While cointegration analysis and correlation measures may appear similar in spirit, they capture distinct aspects of exchange rate dependence. Cointegration tests whether currencies share common stochastic trends, reflecting long-run equilibrium linkages that are stable over time. In contrast, correlations measure the strength of short-run co-movements in returns across different horizons, regardless of whether such movements are anchored by an equilibrium relation. It is therefore possible for currencies to be highly correlated on average without being cointegrated, or conversely, to exhibit cointegration without strong short-run correlation. By employing both approaches, we obtain a richer characterization of exchange rate interdependence that distinguishes structural long-run integration from episodic short-run dependence, particularly across different crisis regimes.

7 Results

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8 Robustness

9 Conclusion

References

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Based on one single stochastic trend/factor driving all, but there are several

10 Appendix: Complete Proofs

Proof of Proposition 1 (Cointegration rank $N - 1$)

For any $b \in \mathbb{R}^N$, note:

$$b'X_t = b'\mu_r + (b'\Lambda_r)g_t + b'\Gamma z_t + b'u_t.$$

Since z_t and u_t are stationary, $b'\Gamma z_t + b'u_t$ is stationary $I(0)$. The only potentially nonstationary component is $(b'\Lambda_r)g_t$, because g_t is $I(1)$ and its components are assumed not to be cointegrated (so they represent independent stochastic trends). Thus:

- If $b'\Lambda_r = 0$, then $b'X_t$ is a constant plus stationary variables $I(0)$ terms $\Rightarrow I(0)$.
- If $b'\Lambda_r \neq 0$, then $b'X_t$ inherits an $I(1)$ component $\Rightarrow I(1)$.

Hence, the cointegrating space is $\{b \in \mathbb{R}^N : b'\Lambda_r = 0\}$, of dimension $N - 1$ when $q_r = 1$ and $\Lambda_r \neq 0$.

(Cointegration rank $N - \text{rank}(\Lambda_r)$)

Given regime r and assumptions A1–A4, the cointegration rank of X_t is:

$$\text{Cointegration rank of } X_t = N - \text{rank}(\Lambda_r).$$

Proof. Consider the long-run factor structure:

$$X_t = \mu + \Lambda_r g_t + \nu_t,$$

where:

- $X_t \in \mathbb{R}^N$ is a vector of exchange rates (e.g., log bilateral exchange rates),
- $\Lambda_r \in \mathbb{R}^{N \times K}$ is the matrix of factor loadings,
- $g_t \in \mathbb{R}^K$ is a vector of global stochastic trends, assumed to be integrated of order one ($I(1)$),
- ν_t is a stationary vector process.

By assumption A2, the components of g_t are mutually uncorrelated and not cointegrated. Thus, g_t represents K independent stochastic trends. The nonstationarity in X_t arises only from the term $\Lambda_r g_t$. Let $r := \text{rank}(\Lambda_r) \leq \min(N, K)$. Then the image of g_t under Λ_r spans an r -dimensional subspace of $\mathbb{R}^N$. This implies that the nonstationary component of X_t lies entirely within an r -dimensional subspace. Let us now consider the existence of cointegrating vectors. A cointegrating vector $\beta \in \mathbb{R}^N$ is such that the linear combination $\beta^\top X_t$ is stationary:

$$\beta^\top X_t = \beta^\top \mu + \beta^\top \Lambda_r g_t + \beta^\top \nu_t.$$

Since μ is constant and ν_t is stationary, the stationarity of $\beta^\top X_t$ depends entirely on whether $\beta^\top \Lambda_r g_t$ is stationary. But g_t is $I(1)$, so $\beta^\top \Lambda_r g_t$ is stationary if and only if:

$$\beta^\top \Lambda_r = 0.$$

That is, cointegrating vectors lie in the left null space of Λ_r . The dimension of this null space is:

$$\dim(\{\beta \in \mathbb{R}^N : \beta^\top \Lambda_r = 0\}) = N - \text{rank}(\Lambda_r).$$

Hence, the number of linearly independent cointegrating vectors—i.e., the cointegration rank of X_t —is:

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

Proposition 2 (Crisis Co-Movement Amplification of Long Horizon Currency Returns)

We begin with the factor model for currency i in regime r :

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top z_t + u_{i,t},$$

where g_t are global stochastic trends (e.g., integrated), z_t are stationary domestic/regional factors, and $u_{i,t}$ is idiosyncratic noise.

Taking first differences:

$$\Delta X_{i,t} = \lambda_{i,r}^\top \Delta g_t + \Delta(\Gamma_i^\top z_t + u_{i,t}).$$

Define the cumulative return over horizon H as:

$$Y_{i,H} \equiv \sum_{h=1}^H \Delta X_{i,t+h} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H},$$

where $\varepsilon_{i,H} = \sum_{h=1}^H \Delta(\Gamma_i^\top z_{t+h} + u_{i,t+h})$ is a sum of stationary terms.

Variance of the long-run return

Because g_t is a vector of random walks,

$$\text{Var}(g_{t+H} - g_t) = H \Sigma_{g,r},$$

and we assume $\varepsilon_{i,H}$ is uncorrelated with g_t . Then:

$$\text{Var}(Y_{i,H}) \approx H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + \text{Var}(\varepsilon_{i,H}).$$

Since $\varepsilon_{i,H}$ is a sum of stationary shocks:

$$\text{Var}(\varepsilon_{i,H}) \approx H \sigma_{\varepsilon,i}^2.$$

Factor share of total variance

The share of total variance explained by global factors is:

$$\frac{\text{Factor variance}}{\text{Total variance}} = \frac{H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}}{H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + H \sigma_{\varepsilon,i}^2} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}}{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + \sigma_{\varepsilon,i}^2}.$$

As $\Sigma_{g,r}$ increases, this ratio increases. Thus, the common global factors explain a larger fraction of the long-run variance. This is the *variance channel* of co-movement amplification.

Correlation between long-run returns

For any pair i, j , define:

$$\rho_{ij}^{(H)} = \frac{\text{Cov}(Y_{i,H}, Y_{j,H})}{\sqrt{\text{Var}(Y_{i,H}) \text{Var}(Y_{j,H})}}.$$

We have:

$$\begin{aligned} \text{Cov}(Y_{i,H}, Y_{j,H}) &= H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + \text{Cov}(\varepsilon_{i,H}, \varepsilon_{j,H}), \\ \text{Var}(Y_{i,H}) &= H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + H \sigma_{\varepsilon,i}^2, \quad \text{Var}(Y_{j,H}) = H \lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + H \sigma_{\varepsilon,j}^2. \end{aligned}$$

Divide numerator and denominator by H :

$$\rho_{ij}^{(H)} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + \frac{1}{H} \text{Cov}(\varepsilon_{i,H}, \varepsilon_{j,H})}{\sqrt{(\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + \frac{1}{H} \sigma_{\varepsilon,i}^2) (\lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + \frac{1}{H} \sigma_{\varepsilon,j}^2)}}.$$

As $H \rightarrow \infty$, the idiosyncratic components vanish:

$$\lim_{H \rightarrow \infty} \rho_{ij}^{(H)} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r}}{\sqrt{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} \cdot \lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r}}}.$$

This is the cosine of the angle between the vectors $\Sigma_{g,r}^{1/2} \lambda_{i,r}$ and $\Sigma_{g,r}^{1/2} \lambda_{j,r}$.

If the factor loadings are aligned (i.e., $\lambda_{i,r} \approx \lambda_{j,r}$), then:

$$\lim_{H \rightarrow \infty} \rho_{ij}^{(H)} \rightarrow 1.$$

Thus, greater similarity in how currencies respond to global shocks leads to stronger long-run correlation. This is the *loading-alignment channel*.

Conclusion

During crises, increases in global factor volatility and convergence in cross-currency loadings jointly amplify long-run return correlations and the dominance of common shocks. This explains tighter global alignment in currency returns during episodes of global financial stress. ■

Proposition 3 (Rank Change in Crisis and Cointegration)

Consider the factor structure of currency returns before and during crisis regimes:

$$X_t = \mu_r + \Lambda_r g_t + \Gamma_r z_t + u_t,$$

where:

- X_t is the $N \times 1$ vector of currency returns,
- Λ_r is the $N \times K_r$ loading matrix on persistent global factors g_t ,

- g_t is a $K_r \times 1$ vector of persistent global factors (stochastic trends),
- $\Gamma_r z_t + u_t$ is a stationary component,
- The regime $r \in \{\text{pre-crisis, crisis}\}$ indexes different periods.

Relation between rank of Λ_r and cointegration

By standard results in cointegration theory (see Johansen 1991; Stock and Watson 1988), the number of common stochastic trends driving X_t equals the rank of Λ_r , denoted $\text{rank}(\Lambda_r) = K_r$. The dimension of the cointegrating space, i.e., the number of stationary linear combinations of X_t , equals

$$\dim(\text{cointegrating space}) = N - K_r.$$

Effect of new persistent global factors

Suppose that during crisis, new persistent global factors emerge that are independent of the pre-crisis factors. Formally, the crisis regime factor vector can be written as

$$g_t^{\text{crisis}} = \begin{bmatrix} g_t^{\text{pre}} \\ g_t^{\text{new}} \end{bmatrix},$$

where $\dim(g_t^{\text{new}}) = K_{\text{new}} > 0$, and g_t^{new} is independent of g_t^{pre} . The corresponding loading matrix expands as

$$\Lambda_{\text{crisis}} = [\Lambda_{\text{pre}} \quad \Lambda_{\text{new}}],$$

with Λ_{new} loading persistently on the new factors. Because g_t^{new} is independent and Λ_{new} has full column rank K_{new} , the rank of Λ_{crisis} satisfies

$$\text{rank}(\Lambda_{\text{crisis}}) \geq \text{rank}(\Lambda_{\text{pre}}) + K_{\text{new}}.$$

Thus,

$$K_{\text{crisis}} = \text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}) = K_{\text{pre}}.$$

Effect on cointegration relations

Since the number of cointegrating relations equals $N - K_r$, an increase in the rank K_r reduces the cointegration dimension:

$$N - K_{\text{crisis}} < N - K_{\text{pre}}.$$

In the extreme case where

$$K_{\text{crisis}} = N,$$

the number of cointegrating relations becomes zero, meaning no stationary linear combination exists — the system has no cointegration.

Effect of autonomy shifts on rank

Alternatively, even without new factors, if autonomy shifts such that the loading matrix Λ changes rank (e.g., if some currency responses become linearly independent from others), then

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

and the same conclusion on cointegration applies.

Conclusion: The emergence of new persistent global factors or autonomy shifts that increase the rank of Λ during crisis reduce the number of cointegrating relations, and if the rank reaches N , cointegration disappears entirely. $\square$

Derivation of (7) from the SDF

Start from the key FX restriction:

$$1 = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right].$$

Substitute $\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r g_{t+1} - v_{t+1}$, express $X_{i,t+1}$ linearly in g_{t+1} plus stationary variables. Under joint normality, only a linear loading on g_t balances the term involving g_{t+1} . Solving yields:

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r} g_t + \Gamma'_i z_t + u_{i,t},$$

with constants capturing Jensen corrections and stationary components as described. Q.E.D.

Deriving the affine form in (18)

Start from the FX restriction

$$1 = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right],$$

with $\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}$, $\xi_{t+1} = \lambda_r g_{t+1} + v_{t+1}$, $g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}$, and $v_{t+1} \perp \varepsilon_{t+1}$, jointly normal. Conjecture (to be verified) that $X_{i,t} = A_{i,r} + B_{i,r} g_t + \Gamma'_i z_t + u_{i,t}$ with $z_t, u_{i,t}$ stationary and independent of ε_{t+1} . Then

$$\log(M_{t+1} e^{X_{i,t} - X_{i,t+1}}) = -r_t - \frac{1}{2}\varsigma_r^2 + B_{i,r} g_t - (B_{i,r} + \lambda_r) g_{t+1} - v_{t+1} - \Gamma'_i (z_{t+1} - z_t) - (u_{i,t+1} - u_{i,t}).$$

Since the shocks are jointly normal and enter linearly, the conditional expectation is the exponential of the conditional mean plus one half the conditional variance. Writing $g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}$, the deterministic (state-dependent) part contains $B_{i,r} g_t - (B_{i,r} + \lambda_r)(g_t + \mu_r) = -\lambda_r g_t - (B_{i,r} + \lambda_r) \mu_r$. For the restriction to hold for all states (g_t, z_t) , the coefficient on g_t must vanish. Hence $B_{i,r} = \lambda_{i,r}$. All remaining constants and stationary terms are absorbed into $\mu_{i,r}$, Γ_i , and $u_{i,t}$, yielding the affine solution $X_{i,t} = \mu_{i,r} + \lambda_{i,r} g_t + \Gamma'_i z_t + u_{i,t}$.

Proposition (Affine representation of $X_{i,t}$)

Under regime r , suppose (i) the SDF is log-normal,

$$\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad \xi_{t+1} = \lambda_r g_{t+1} + v_{t+1},$$

with $\varsigma_r^2 = \lambda_r^2 \sigma_{g,r}^2 + \sigma_v^2$; (ii) the global factor follows a random walk $g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}$; (iii) ε_{t+1} and v_{t+1} are conditionally Gaussian and independent; (iv) the $K \times 1$ fundamentals z_t are stationary and independent of ε_{t+1} , and the stationary block $\Gamma'_i z_t + u_{i,t}$ has conditionally mean-zero first differences and finite variance.⁴ Then the no-arbitrage FX restriction

$$1 = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right] \tag{17}$$

admits a log-linear (affine) solution

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \quad u_{i,t} \text{ stationary}, \tag{18}$$

⁴Any nonzero *constant* drift in the stationary block can be absorbed into the intercept below.

with $\lambda_{i,r}$ a currency- and regime-specific loading on the single nonstationary factor g_t , and

$$\mu_{i,r} = -\bar{r}_r - (\lambda_{i,r} + \lambda_r) \mu_r + \frac{1}{2} [(\lambda_{i,r} + \lambda_r)^2 \sigma_{g,r}^2 + \sigma_v^2 + (\Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}))], \quad (19)$$

where $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$. Moreover, if $X_{i,t}$ had any additional I(1) component orthogonal to g_t , (17) would depend on that state and fail to hold for all $(g_t, \mathbf{z}_t)$; hence g_t spans the only nonstationary direction. Guess (18). Then

$$X_{i,t} - X_{i,t+1} = -\lambda_{i,r}(g_{t+1} - g_t) - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}) = -\lambda_{i,r}\mu_r - \lambda_{i,r}\sigma_{g,r}\varepsilon_{t+1} - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}).$$

Stack the innovations $\eta_{t+1} \equiv (\varepsilon_{t+1}, v_{t+1}, \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}))'$. By assumptions, $\eta_{t+1} \mid \mathcal{F}_t$ is (jointly) normal with mean zero and finite covariance, and $Y_{t+1} \equiv \log M_{t+1} + X_{i,t} - X_{i,t+1}$ is *linear* in η_{t+1} :

$$Y_{t+1} = -r_t - \frac{1}{2}\zeta_r^2 - \lambda_r g_{t+1} + \lambda_{i,r} g_t - \lambda_{i,r} \mu_r - v_{t+1} - \lambda_{i,r} \sigma_{g,r} \varepsilon_{t+1} - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}).$$

Using $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$ gives

$$Y_{t+1} = \underbrace{-r_t - \frac{1}{2}\zeta_r^2 - \lambda_r g_t - (\lambda_{i,r} + \lambda_r) \mu_r}_{\text{deterministic given } \mathcal{F}_t} + \underbrace{-(\lambda_{i,r} + \lambda_r) \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t})}_{\text{linear in Gaussian shocks}}.$$

Since $Y_{t+1} \mid \mathcal{F}_t$ is normal,

$$\mathbb{E}_t[e^{Y_{t+1}}] = \exp\left(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2}_t(Y_{t+1})\right).$$

Imposing (17) therefore requires

$$\mathbb{E}_t[Y_{t+1}] + \frac{1}{2}_t(Y_{t+1}) = 0.$$

By the display above,

$$\mathbb{E}_t[Y_{t+1}] = -r_t - \frac{1}{2}\zeta_r^2 - \lambda_r g_t - (\lambda_{i,r} + \lambda_r) \mu_r,$$

$$\frac{1}{2}_t(Y_{t+1}) = \frac{1}{2} [(\lambda_{i,r} + \lambda_r)^2 \sigma_{g,r}^2 + \sigma_v^2 + (\Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}))].$$

For $\mathbb{E}_t[Y_{t+1}] + \frac{1}{2}_t(Y_{t+1})$ to be *state-independent* (hold for all $(g_t, \mathbf{z}_t)$), any g_t -dependence in r_t must cancel the $-\lambda_r g_t$ term; this is the standard affine restriction on short rates.⁵ Taking

⁵Allow the short rate to be affine in the state, $r_t = r_{0,r} + r_{g,r} g_t + r'_{z,r} \mathbf{z}_t$; the identity for all g_t implies $r_{g,r} = -\lambda_r$ and $r_{z,r} = 0$. This matches the “prices are exponential-affine in states” logic of affine pricing.

expectations across r_t within the regime and collecting constants then yields (19). Finally, if $X_{i,t}$ contained any I(1) direction orthogonal to g_t , it would enter Y_{t+1} without a matching term in ξ_{t+1} , making $\mathbb{E}_t[e^{Y_{t+1}}]$ depend on that state and violating (17). Hence g_t is the *only* nonstationary driver, and (18) follows.

Start from the SDF SDE

Assume the stochastic discount factor M_t follows

$$\frac{dM_t}{M_t} = -r_t dt - \Lambda_t^\top dW_t, \quad (20)$$

where r_t is the short rate, W_t is a k -dimensional Brownian motion, and $\Lambda_t \in \mathbb{R}^k$ collects the (vector) prices of risk.

Itô's lemma for $\log M_t$

By Itô's formula,

$$d \log M_t = \frac{dM_t}{M_t} - \frac{1}{2} \left(\frac{dM_t}{M_t} \right)^2.$$

Using (20), note that $\left(\frac{dM_t}{M_t} \right)^2 = (\Lambda_t^\top dW_t)^2 = \|\Lambda_t\|^2 dt$. Hence

$$d \log M_t = -r_t dt - \Lambda_t^\top dW_t - \frac{1}{2} \|\Lambda_t\|^2 dt. \quad (21)$$

Integrate over a finite interval

Integrate (21) from t to $t + \Delta$:

$$\log M_{t+\Delta} - \log M_t = - \int_t^{t+\Delta} r_s ds - \frac{1}{2} \int_t^{t+\Delta} \|\Lambda_s\|^2 ds - \int_t^{t+\Delta} \Lambda_s^\top dW_s. \quad (22)$$

Define the (martingale) shock and its conditional variance:

$$\xi_{t+\Delta} \equiv \int_t^{t+\Delta} \Lambda_s^\top dW_s, \quad \varsigma^2 \equiv \int_t^{t+\Delta} \|\Lambda_s\|^2 ds.$$

Conditionally on $\mathcal{F}_t$, $\xi_{t+\Delta} \sim \mathcal{N}(0, \varsigma^2)$. Then (22) becomes

$$\log M_{t+\Delta} = \log M_t - \int_t^{t+\Delta} r_s ds - \frac{1}{2} \varsigma^2 - \xi_{t+\Delta}. \quad (23)$$

One-period, piecewise-constant specialization

If we take $\Delta = 1$ and assume $r_s \equiv r_t$ and $\Lambda_s \equiv \Lambda_t$ on $[t, t + 1]$, then

$$\int_t^{t+1} r_s ds = r_t, \quad \varsigma^2 = \int_t^{t+1} \|\Lambda_s\|^2 ds = \|\Lambda_t\|^2 \equiv \varsigma_r^2,$$

and $\xi_{t+1} = \Lambda_t^\top (W_{t+1} - W_t)$. Normalizing $M_t = 1$ (so $\log M_t = 0$), (23) reduces to

$$\boxed{\log M_{t+1} = -r_t - \frac{1}{2} \varsigma_r^2 - \xi_{t+1}}, \quad (24)$$

with $\xi_{t+1} | \mathcal{F}_t \sim \mathcal{N}(0, \varsigma_r^2)$. Since $\xi_{t+1} \sim \mathcal{N}(0, \varsigma_r^2)$, $\mathbb{E}_t[e^{-\xi_{t+1}}] = \exp(\frac{1}{2} \varsigma_r^2)$. Therefore

$$\mathbb{E}_t[M_{t+1}] = \exp\left(-r_t - \frac{1}{2} \varsigma_r^2\right) \mathbb{E}_t[e^{-\xi_{t+1}}] = \exp(-r_t),$$

so $-\frac{1}{2} \varsigma_r^2$ is exactly the Jensen correction that pins $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

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